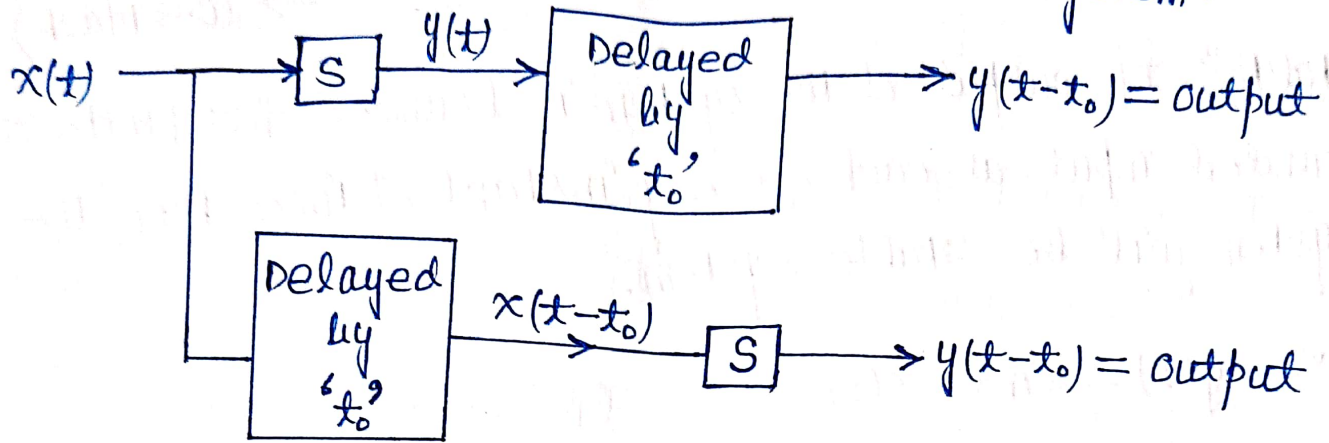


Solution 1(a) :-

Time invariant :- Any delay provided at input must be reflected in output for Time invariant system.

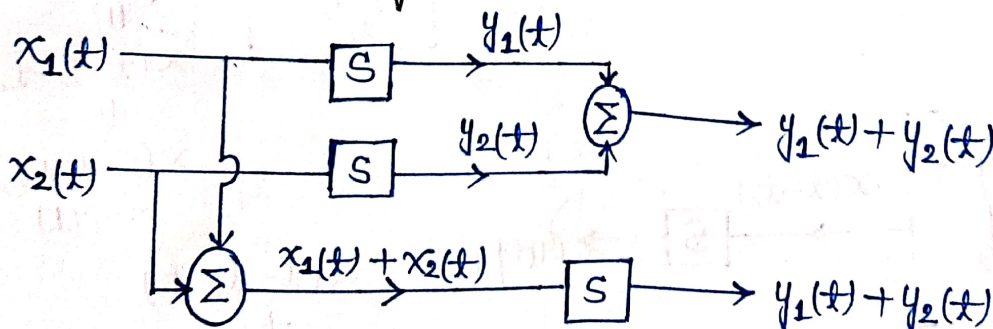


Ex: $y(t) = x(\sin t)$

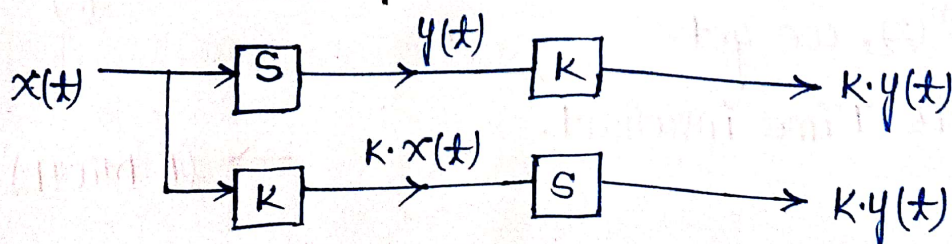
→ (0.5 Mark)

Linear :- A linear system follows the law of superposition. It is a combination of two laws - (I) Law of additivity (II) Law of Homogeneity Ex: $y(t) = \cos t \cdot x(t)$

(I) Law of additivity



(II) Law of Homogeneity



→ (0.5 Mark)

Causal ÷ If output of the system is independent of future value of input at each & every instant of time then system will be causal system.

Ex: $y(t) = x(t) + x(t-1)$

→ (0.5 Mark)

Stable ÷ If output of the system is bounded for finite or bounded input at each & every instant of time then the system will be stable system.

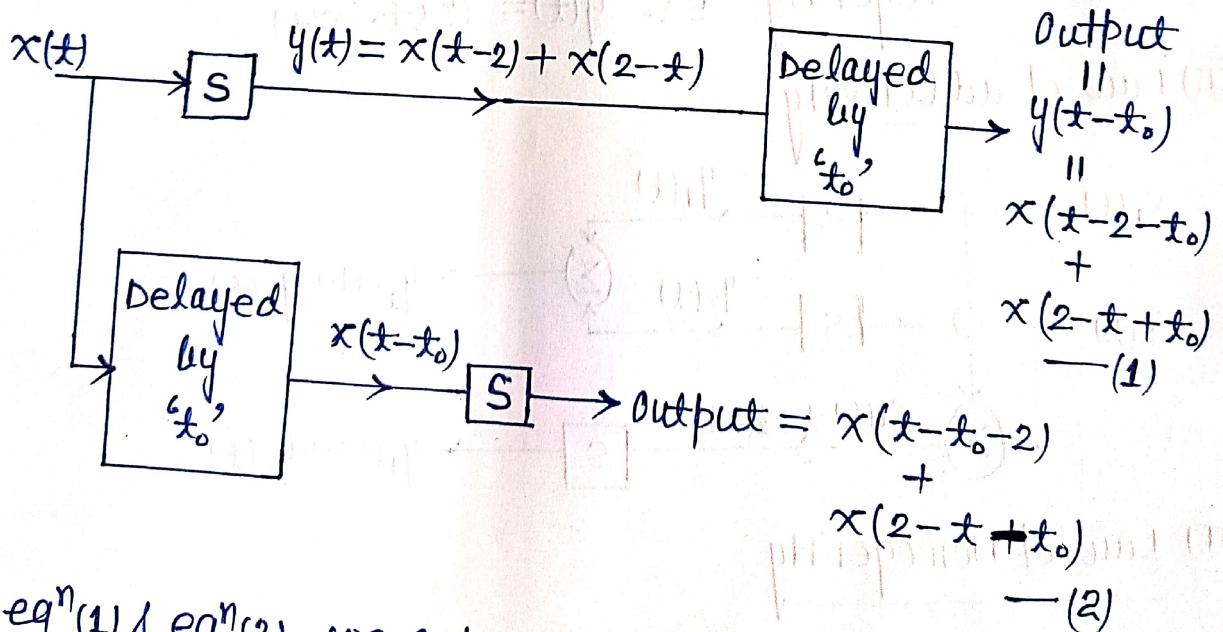
Ex: $y(t) = \sin t \cdot x(t)$

→ (0.5 Mark)

Solution 1(b) ÷

(I) Given— $y(t) = x(t-2) + x(2-t)$

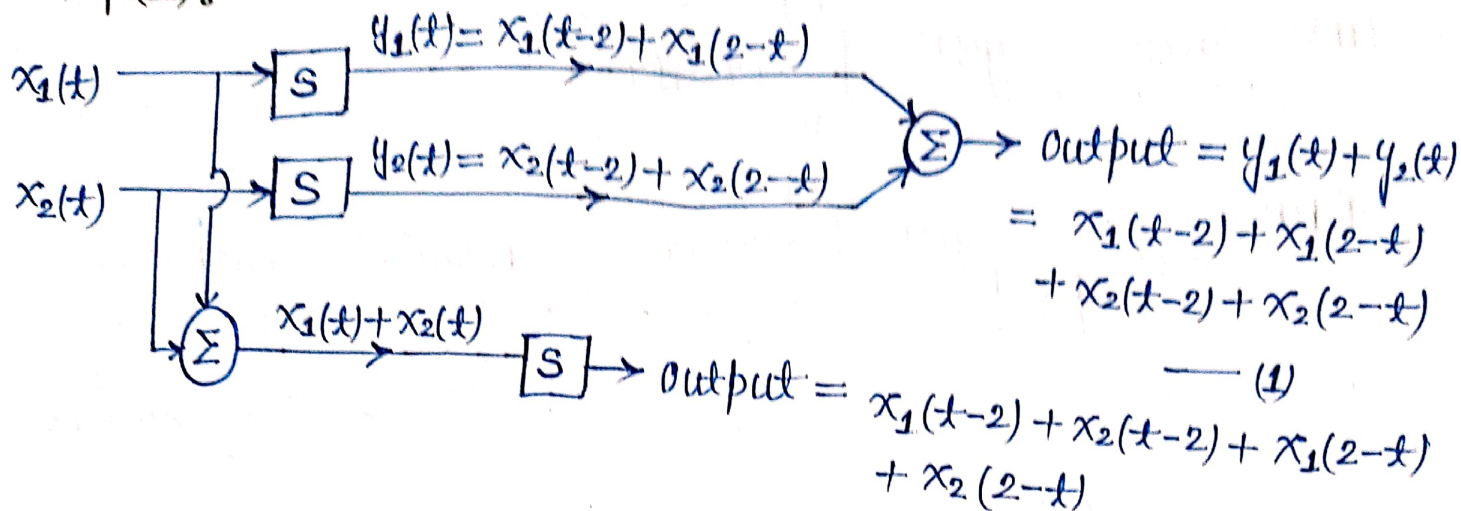
step(1) ÷



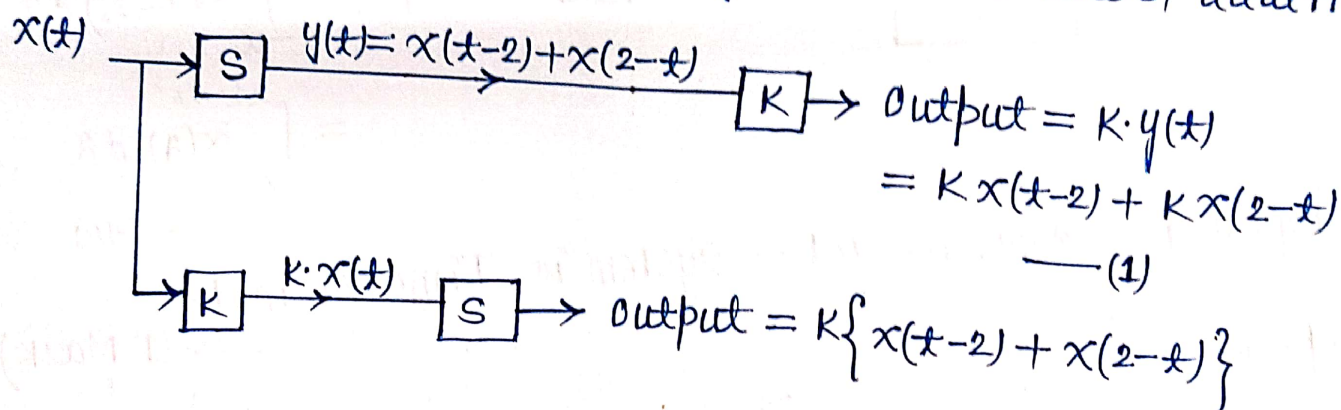
By eqⁿ(1) & eqⁿ(2), we get—
system is Time ~~in~~variant.

→ (1 Mark)

Step (II) ÷



By eqⁿ(1) & eqⁿ(2), we get — System follow law of additivity



By eqⁿ(1) & eqⁿ(2), we get — System follow law of Homogeneity.
Hence, System is Linear.

→ (1 Mark)

Step (3) ÷ Let's check at, $t=0$

$$y(t) = x(+2) + x(-2)$$

$\downarrow$ $\downarrow$
 Future Past
 Value Value

Hence, System is Non-Causal.

→ (1 Mark)

Step (4) ÷ System is Stable

Let, $|x(t)| \leq M$

$\therefore |y(t)| = |x(t-2) + x(2-t)| \leq 2M$

Output will always a finite value.

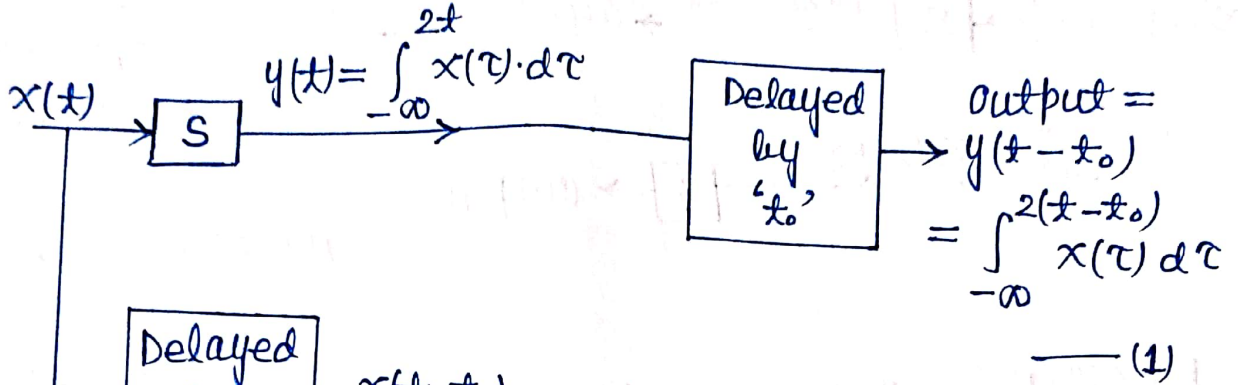
→ (1 Mark)

Solution 1(b)÷

(II) Given-

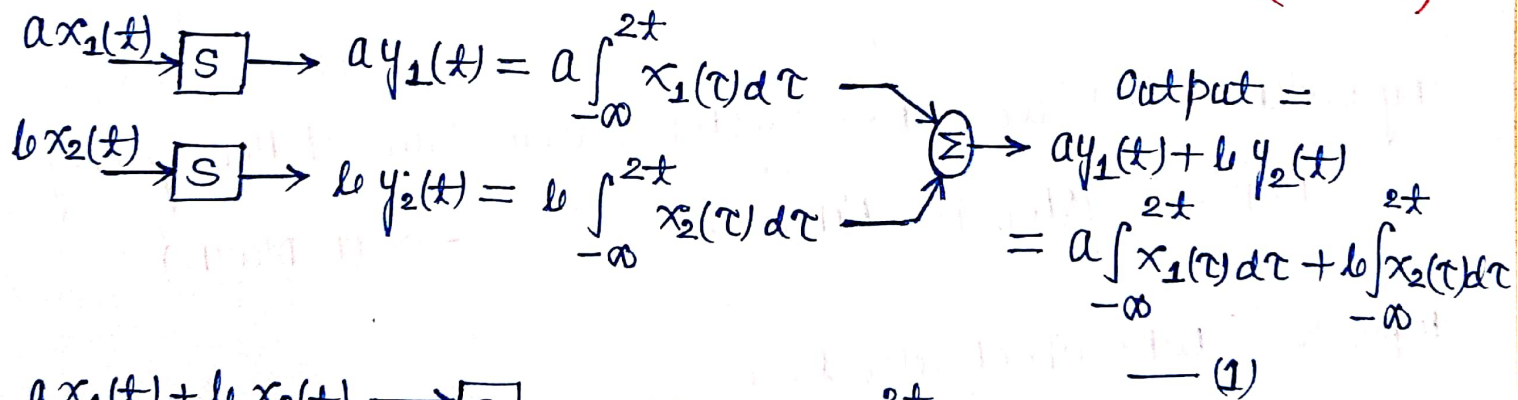
$$y(t) = \int_{-\infty}^{2t} x(\tau) \cdot d\tau$$

Step(1)÷



By eqⁿ(1) & eqⁿ(2), we get - System is Time Variant. —(2)

Step(2)÷



$$a x_1(t) + b x_2(t) \rightarrow [S] \rightarrow \text{Output} = \int_{-\infty}^{2t} \{a x_1(\tau) + b x_2(\tau)\} d\tau$$

—(2)

By eqⁿ(1) & eqⁿ(2), we get - System is Linear. → (1 Mark)

Step(3)÷ At time ' t ', upper limit is ' $2t$ ', so it depends on future value of $x(t)$ if $2t > t$. Hence, System is Non-causal. → (1 Mark)

step(4)÷ Let's assume, $|x(t)| \leq M$ then -

$$|y(t)| = \left| \int_{-\infty}^{2t} x(\tau) d\tau \right| \leq \int_{-\infty}^{2t} |x(\tau)| d\tau$$

This integration over an infinite interval, which can diverge. So even if input is bounded, the output can be unbounded.

Hence, system is Not-stable.

→ (1 MARK)